

Test Report - VxMark Cold Storage

Author: Pius Wong , created Jun 8, 2026 , updated Jun 17, 2026

Summary

The VxMark ballot-marking device (BMD) showed no loss in function from cold storage at -4°F for 78 hours after temperature stabilization and then gradual return to room temperature.

Applicable VVSG Requirements

- 2.7-D – Ability to support maintenance and repair physical environment conditions – non-operating
- 2.7-E – Ability to support transport and storage physical environment conditions – non-operating
- 2.7-F – Ability to support storage temperatures in physical environment – non-operating

Devices Under Test

- VxMark, v4.1, image v4.0.4

Purpose

VxMark must resist cold storage at low temperature according to VVSG 2.7-F down to -4°F. It was not expected to be vulnerable to cold storage because (a) its components are rated for this cold storage temperature, and (b) VxScan previously passed cold storage testing and shares many components with VxMark as documented in *Test Report - VxScan Cold Storage*. Testing was done internally to confirm this. Procedures did not have to be tightly controlled as in an NRTL due to the low predicted risk of failure.

Materials

- VxMark unit (MK-14-018)
 - Imaged with v4.0.4 production app
 - Configured with summary ballot test election
- VxMark printer (HP4001dn, VNL0728678) and paper
- Goldenmate 1500VA/1000W UPS (JUO2HE390707)
- VxScan unit for verifying printed ballots

- USB stick with election definition
- Smart cards configured for election
- Cold climate-control space
 - Freezer (Insignia 14 cu. ft.)
 - Temperature controller thermostat (Inkbird WiFi ITC-308)
 - Plastic wrap, bubble wrap, tape, supporting material
 - Smart thermometer-hygrometer (Govee H5075)

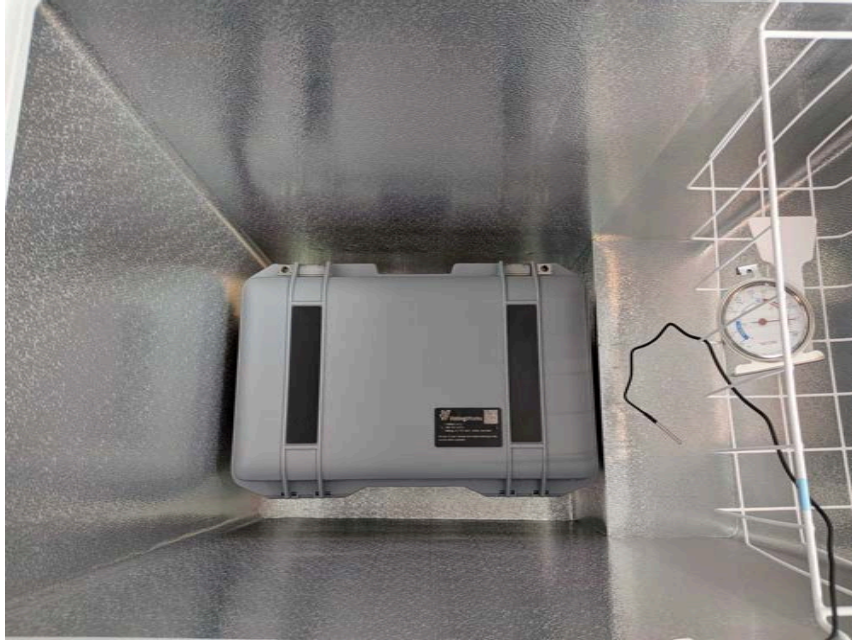
Procedures

1. Check the VxMark BMD for normal function before cold storage.
 - a. Follow tests of normal function (*VxMark Tests of Normal Function*) to confirm the test unit functions normally before any disruptive tests.
 - b. Document tests of normal function. Record any pre-existing problems. Use the pre-existing template.
2. Place the VxMark in the cold climate-control space and set to -4°F. Plan for storage for at least 72 hours, or 3 days, after temperature stabilization.
3. After the time has passed, turn off climate control and let the unit come back up to room temperature at a natural rate to prevent excess condensation.
4. Check the VxMark unit for normal function after cold storage.

Results

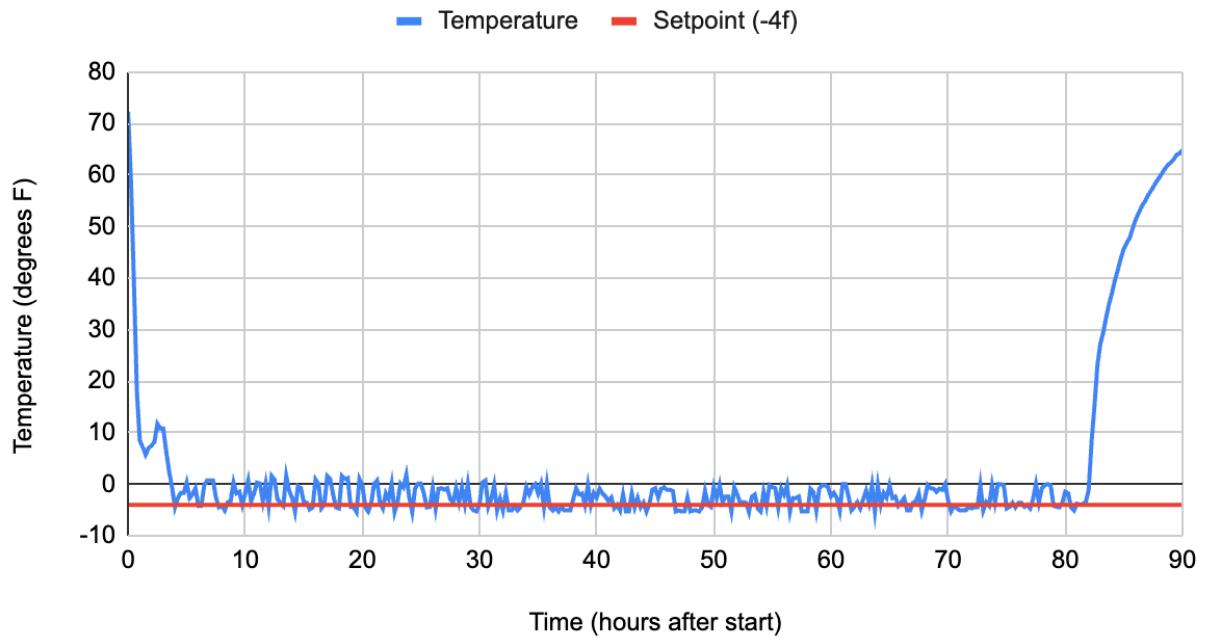
Normal function of the VxMark BMD was confirmed before testing, as documented in [Appendix A](#).

The VxMark was set up to experience controlled cold storage by incorporating the Inkbird temperature controller. The setup is pictured below, where the temperature probe was inserted into the freezer above the test unit. The freezer itself was left closed during testing:



The VxMark unit experienced cold storage for 78 hours after temperature stabilization targeting -4°F. Because of the lower fidelity nature of the test setup, the actual temperature fluctuated around this setpoint, as shown in the freezer temperature readings over time graphed below.

Storage Temperature vs. Time



Normal function was confirmed after coming back up to temperature from cold storage, as documented in [Appendix B](#).

Conclusions

- The VxMark BMD is likely to pass standard cold storage tests required by VVSG 2.7-F up to 78 hours.
- Future cold storage testing should account for the error in reaching the temperature setpoint and lower the temperature controls.

Appendix A: VxMark Normal Function Checklist - 20260603 - before cold

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong, VotingWorks, Austin	
	Date & time tested	6/3/2026	11:40 AM
	System tested & Unit/Model No.:	VxMark: MK-14-018	
	Before what other test?	Cold storage, unpackaged, 72-hour	
	After what other test?	Vibration & regular storage	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	PASS	
2	Visually displays instructions and accurate feedback to users.	PASS	
3	Audibly plays instructions and accurate feedback to users.	PASS	
4	Accepts user inputs on the touchscreen.	PASS	
5	Accepts user inputs on the ATI & PAT	PASS	
6	Accepts inputs on the barcode scanner.	PASS	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			

No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	PASS	
8	Protects internal electronics and subsystems.	PASS	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	PASS	
2	Touchscreen	PASS	
3	Power Supply, internal	n/a	not opened up to inspect
4	Smart Card Reader	PASS	
5	USB-A receptacle	PASS	
6	USB 3 Switch	n/a	not opened up to inspect
7	Barcode Scanner	PASS	
8	Cable Assemblies	n/a	not opened up to inspect
9	Uninterruptible Power Supply (external to unit)	n/a	not used in this test
10	Printer (external to unit)	PASS	normal print, small specks of ink/toner on left part of page, could clean later
11	Outer case	PASS	

12	Inner panels around electronics	n/a	not opened up to inspect
13	USB cover	PASS	
14	Security tie points	PASS	
15	Fasteners, both external and internal	PASS	not opened up to inspect
16	Wires, ports, and electrical connections	PASS	not opened up to inspect
18	other....		

Appendix B: VxMark Normal Function Checklist - 20260608 - after cold

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong, VotingWorks, Austin	
	Date & time tested	6/8/2026	10:29 AM
	System tested & Unit/Model No.:	VxMark: MK-14-018	
	Before what other test?	n/a	
	After what other test?	Cold storage, unpackaged, 72-hour	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	PASS	
2	Visually displays instructions and accurate feedback to users.	PASS	
3	Audibly plays instructions and accurate feedback to users.	PASS	
4	Accepts user inputs on the touchscreen.	PASS	
5	Accepts user inputs on the ATI & PAT	PASS	sip/puff tested
6	Accepts inputs on the barcode scanner.	PASS	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes

7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	PASS	
8	Protects internal electronics and subsystems.	PASS	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	PASS	
2	Touchscreen	PASS	
3	Power Supply, internal	n/a	not opened up to inspect
4	Smart Card Reader	PASS	
5	USB-A receptacle	PASS	also saved logs successfully to USB
6	USB 3 Switch	n/a	not opened up to inspect
7	Barcode Scanner	PASS	
8	Cable Assemblies	n/a	not opened up to inspect
9	Uninterruptible Power Supply (external to unit)	n/a	not used in this test
10	Printer (external to unit)	PASS	
11	Outer case	PASS	
12	Inner panels around electronics	n/a	not opened up to inspect
13	USB cover	PASS	
14	Security tie points	PASS	
15	Fasteners, both external and internal	PASS	not opened up to inspect
16	Wires, ports, and electrical connections	PASS	not opened up to inspect
18	other....		